Agentic AI Framework in Hydrology

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Automatic Quantification and Unified Agent in Hydrology



Eight Specialized Agents Orchestrate the Complete Workflow

AQUAH implements a multi-agent system where eight specialized agents collaborate to transform user requests into reproducible simulations. Each agent owns a well-defined responsibility and passes structured artifacts to the next, enabling transparent reasoning, easier debugging, and seamless extensibility.

A_{CP} Context Parser Agent

Extracts geographic extent and temporal windows from natural language requests

A_{DR} Dataset Retriever Agent

Fetches and clips all required geospatial layers and forcing data

A_{P} Perceptor Agent

Employs vision-augmented LLMs to interpret maps and extract morphological descriptors

A_{OS} OutletSelector Agent

Applies hydrologic heuristics to identify optimal gauge locations

A_{PI} ParamInitializer Agent

Uses Retrieval-Augmented Generation to propose physically plausible initial parameters

A_{O} Operator Agent

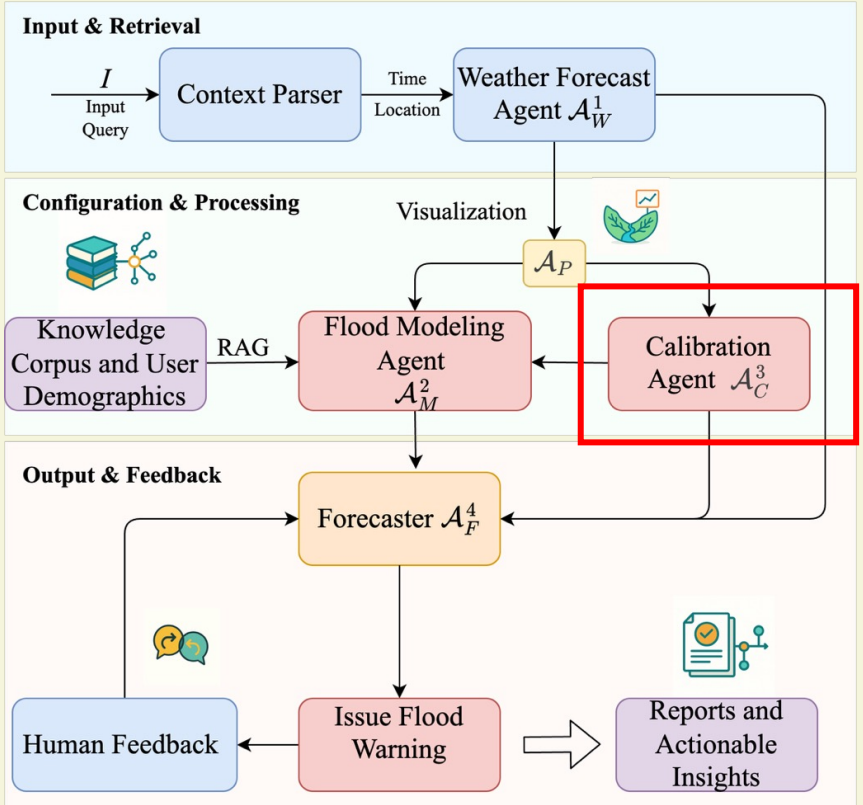
Executes the CREST simulation and captures performance metrics

A_{RW} Report Writer Agent

Compiles comprehensive documentation with maps, hydrographs, and statistics

A_{FR} Feedback Reflector Agent

Processes user commentary to enable iterative refinement



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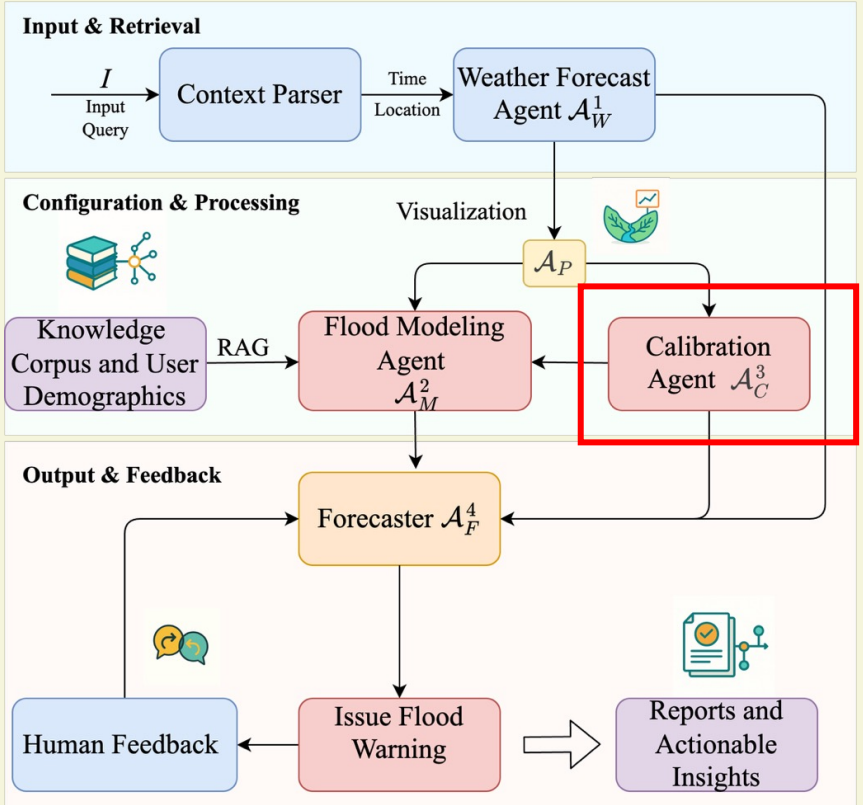
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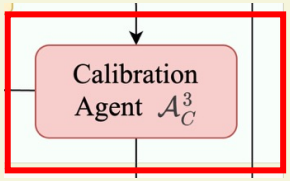
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1. Time Series Comparison



Why using agents for calibrating Earth system models?

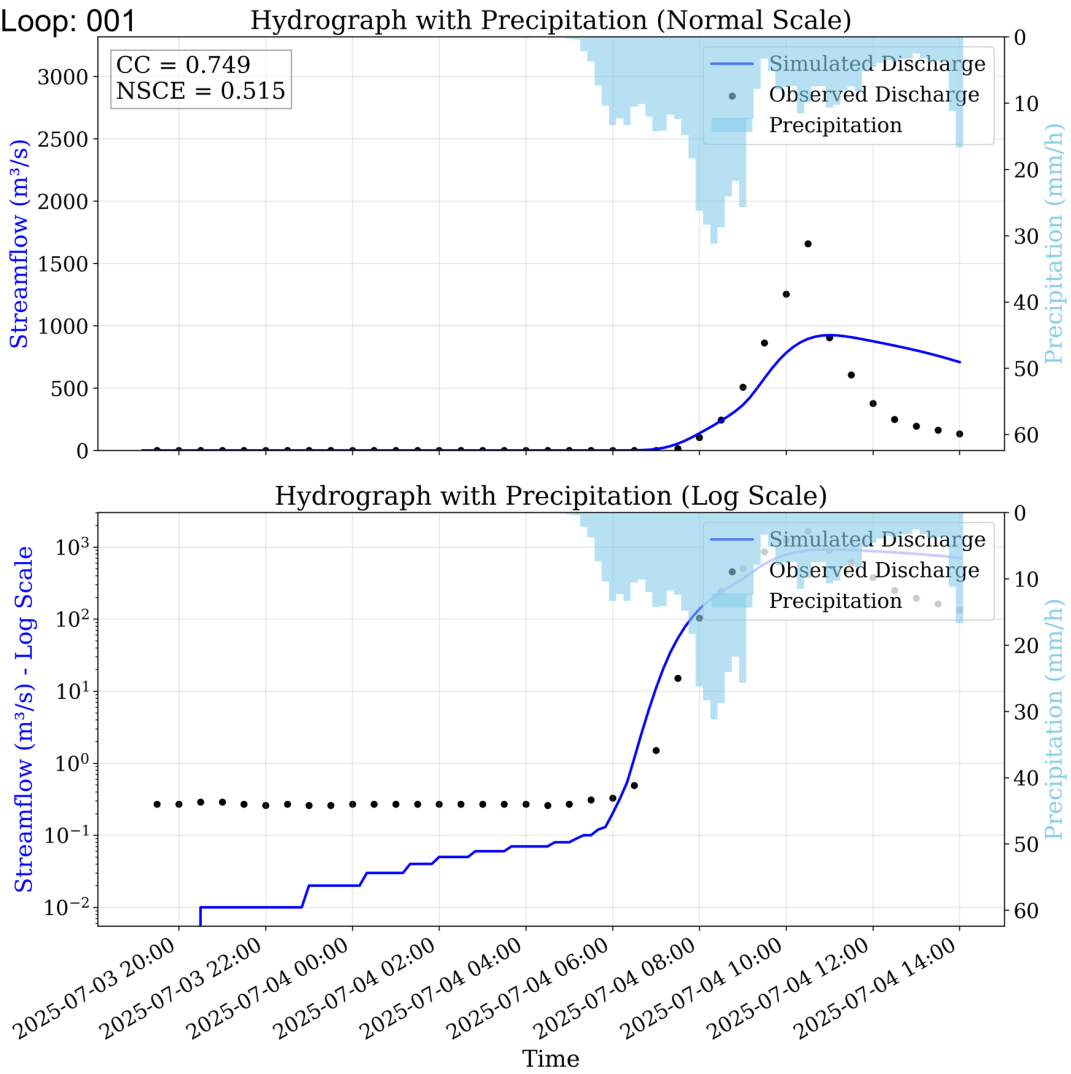
1. Current optimization schemes are stochastic (or in brutal force)
2. It relies on a single metric for calibration

Agent Calibration



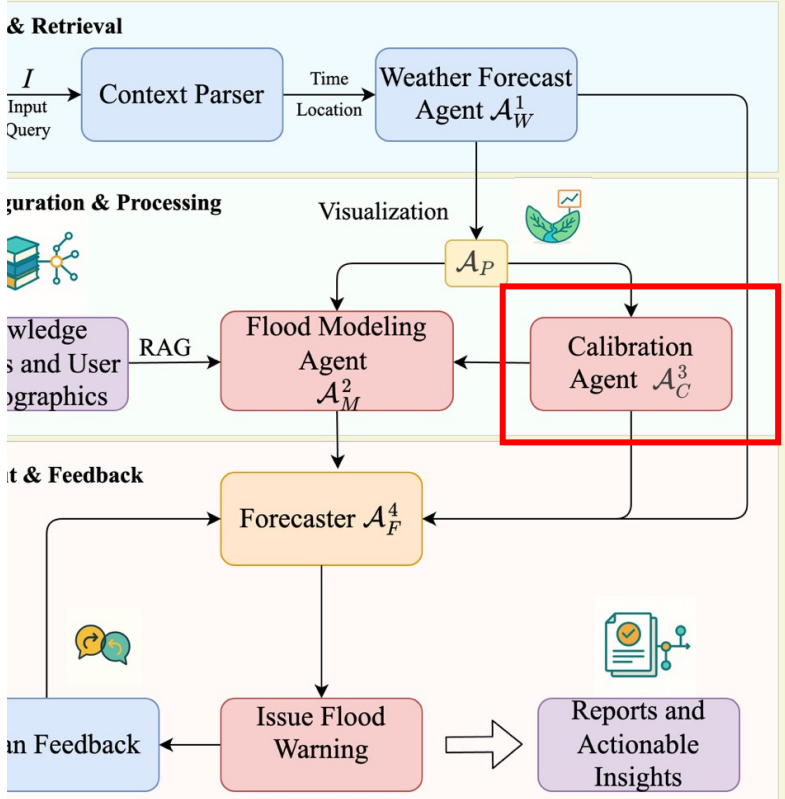
1. Physics-based parameter adjustment
2. Interpreting the whole graph (vision calibration)

Loop: 001



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Working on the last mile challenge

terminal-bench
science

